HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Contributed to **RTL design** and micro-architecture implementation for microcontroller subsystem components, working within simulation environments to validate functionality.
- Developed and maintained **SystemVerilog** code for microcontroller subsystem components, focusing on functional correctness and performance.
- Created **Python scripts** to automate engineering workflows, including data parsing and report generation, improving team efficiency by an estimated **15%**.
- Collaborated with cross-functional teams, including verification and architecture, to analyze and resolve design-related challenges throughout the development lifecycle.
- Participated in **code reviews** for RTL designs, ensuring adherence to coding standards and best practices.
- Assisted in the integration of **C++** components with existing microcontroller subsystems, debugging issues related to interfaces and data flow.
- Managed and deployed containerized applications using **Docker** on **AWS EC2** instances, configuring basic resource limits for predictable performance.

SKILLS

Languages – SystemVerilog, C, C++, Python, Perl

Architecture – CPU Architecture, Micro-architecture

Tools & Environments – RTL Simulation, Jenkins, Git, Jira, Confluence

Cloud & Containerization – AWS, EC2, S3, Lambda, Docker

Scripting – Python, Shell Scripting

PROJECTS

RISC-V Core RTL Development

- Designed and implemented core **RTL modules** for a RISC-V microcontroller subsystem using **SystemVerilog**, focusing on instruction decoding and execution pipelines.
- Integrated custom peripherals with the main CPU core, ensuring proper bus interface and data transfer protocols.
- Developed simulation testbenches in SystemVerilog to verify functional correctness and identify design flaws, reducing bug discovery latency.

Embedded Linux Automation Suite

- Created a suite of **Python scripts** to automate the build, deployment, and testing of embedded Linux firmware for microcontrollers.
- Implemented automated log analysis and error reporting using **shell scripting** and **Python**, reducing manual triage time by **30%** for system integration testing.
- Containerized the build environment using **Docker**, ensuring consistent and reproducible firmware builds across development and testing stages.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)